

1. Domain Architecture & Network Topology

2. DNS Zone Record Configuration (adviceintelligence.tech)

Log in to your DNS provider (e.g. Cloudflare, Google Cloud DNS, AWS Route 53, Namecheap, GoDaddy) for the `adviceintelligence.tech` zone and configure the following records:

Table of DNS Records

[!TIP]

If using Cloudflare DNS, ensure SSL/TLS encryption mode is set to Full (Strict) to guarantee end-to-end TLS 1.3 encryption between Cloudflare edge and the Google Cloud origin.

3. Hosting & Deployment Options

Choose the deployment model that fits your infrastructure:

Option A: Google Cloud Run (Recommended — UK Sovereign europe-west2)

Deploying directly to Google Cloud Run maintains strict europe-west2 sovereignty and provides automated Google-managed SSL certificates.

Step 1: Build & Push Container Images

```
# 1. Set GCP project and region
export GCP_PROJECT_ID="advice-intelligence-prod"
export REGION="europe-west2"
gcloud config set project $GCP_PROJECT_ID
gcloud config set compute/region $REGION

# 2. Authenticate Docker with Google Artifact Registry
gcloud auth configure-docker ${REGION}-docker.pkg.dev

# 3. Build & push Backend container image
docker build -f infrastructure/docker/Dockerfile.backend \
  -t ${REGION}-docker.pkg.dev/${GCP_PROJECT_ID}/case-ace/backend:2.0.0 .
docker push ${REGION}-docker.pkg.dev/${GCP_PROJECT_ID}/case-ace/backend:2.0.0

# 4. Build & push Frontend SPA container image (Nginx hardened)
```

```
docker build -f infrastructure/docker/Dockerfile.client \
  --build-arg VITE_APP_ENV=pilot \
  -t ${REGION}-docker.pkg.dev/${GCP_PROJECT_ID}/case-ace/frontend:2.0.0 .
docker push ${REGION}-docker.pkg.dev/${GCP_PROJECT_ID}/case-ace/frontend:2.0.0
```

Step 2: Deploy Cloud Run Services

```
# Deploy Backend API Service
gcloud run deploy case-ace-api \
  --image ${REGION}-docker.pkg.dev/${GCP_PROJECT_ID}/case-ace/backend:2.0.0 \
  --region ${REGION} \
  --platform managed \
  --allow-unauthenticated \
  --set-env-vars APP_ENV=pilot,GCP_REGION=europe-west2,CORS_ORIGIN="https://
caseace.adviceintelligence.tech,https://api.caseace.adviceintelligence.tech"

# Deploy Frontend SPA Service
gcloud run deploy case-ace-frontend \
  --image ${REGION}-docker.pkg.dev/${GCP_PROJECT_ID}/case-ace/frontend:2.0.0 \
  --region ${REGION} \
  --platform managed \
  --allow-unauthenticated
```

Step 3: Map Custom Domains in Cloud Run

```
# Map Frontend to caseace.adviceintelligence.tech
gcloud beta run domain-mappings create \
  --service case-ace-frontend \
  --domain caseace.adviceintelligence.tech \
  --region ${REGION}

# Map Backend to api.caseace.adviceintelligence.tech
gcloud beta run domain-mappings create \
  --service case-ace-api \
  --domain api.caseace.adviceintelligence.tech \
  --region ${REGION}
```

Cloud Run will display the exact DNS records (typically ghs.googlehosted.com.). Google will automatically provision and renew a managed SSL/TLS certificate within 15–30 minutes of DNS propagation.

Option B: Deploying Frontend via Vercel / Cloudflare Pages

If hosting the frontend on modern edge CDN:

Connect your repository to Vercel or Cloudflare Pages.

Set Root Directory to client/.

Set Build Command to npm run build and Output Directory to dist.

Configure Environment Variable: VITE_APP_ENV=pilot.

Add Custom Domain: caseace.adviceintelligence.tech.

Add rewrite rule in vercel.json or Cloudflare Transform Rules to route /api/* to https://api.caseace.adviceintelligence.tech/:splat.

Option C: Ubuntu Linux VPS / Dedicated NGINX Server

If deploying on a managed Ubuntu VM:

```
# 1. Clone repository on server
git clone https://github.com/adviceintelligence/case-ace-v2.git /opt/case-ace-v2
cd /opt/case-ace-v2

# 2. Configure Environment File
cp infrastructure/env/.env.pilot.example /etc/case-ace/production.env

# 3. Issue SSL Certificate via Let's Encrypt Certbot
sudo apt-get install -y certbot python3-certbot-nginx
sudo certbot certonly --nginx -d caseace.adviceintelligence.tech -d api.caseace.adviceintelligence.tech

# 4. Start Docker Containers
docker compose -f docker-compose.prod.yml up -d --build
```

4. Updating Identity & Telephony Integrations

After provisioning caseace.adviceintelligence.tech, update the external identity and telephony OAuth callbacks:

1. Microsoft Entra ID App Registration

Navigate to the Microsoft Entra Admin Center (<https://entra.microsoft.com>).

Go to Identity → Applications → App registrations → Select Case Ace Consultation Assistant.

Under Authentication → Redirect URIs:

Add Web / SPA Redirect URI:

<https://caseace.adviceintelligence.tech/auth/callback>

Under Single Sign-On, ensure ID tokens and Access tokens are permitted for PKCE authorization flows.

Save changes.

2. Cisco Webex Developer Portal

Navigate to the Webex Developer Portal (<https://developer.webex.com/my-apps>).

Select your Case Ace Telephony Ingest Integration.

Under Redirect URI(s):

Add: <https://caseace.adviceintelligence.tech/api/auth/webex/callback>

Add: <https://api.caseace.adviceintelligence.tech/api/auth/webex/callback>

Save changes.

5. Post-Deployment Verification & Smoke Test

Run the following verification suite from your terminal once DNS propagation is complete:

```
# 1. Test HTTPS & TLS 1.3 Handshake on Frontend
curl -s -I https://caseace.adviceintelligence.tech | grep -iE "http|content-security-policy|strict-transport-security"

# Expected Output:
# HTTP/2 200 (or HTTP/1.1 200 OK)
# strict-transport-security: max-age=63072000; includeSubDomains; preload
# content-security-policy: default-src 'none'; script-src 'self' 'wasm-unsafe-eval'...

# 2. Test Backend Health Check & Cloud Sovereignty
curl -s https://api.caseace.adviceintelligence.tech/health

# Expected Output:
# {"status":"healthy","gcpRegion":"europe-west2","version":"2.0.0"}

# 3. Test Ephemeral Credential Minting Endpoint (Authenticated)
curl -s -X POST https://api.caseace.adviceintelligence.tech/api/v1/credentials/issue \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <ADVISER_JWT>" \
  -d '{"purpose":"speech-to-text"}'

# Expected Output:
# {"accessToken":"gcp_sts_...", "expiresInSeconds":300, "region":"europe-west2"}
```

6. Ongoing Maintenance & Health Monitoring

SSL Expiry Alerting: Automated monitoring via uptime check probes testing TLS validity every 24 hours.

Downtime Fallback: If caseace.adviceintelligence.tech is unreachable, advisers immediately execute the Business Continuity Procedure and switch to manual note-taking in Casebook CRM. Advice delivery is never delayed.